

Shangbin Feng

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Last updated: July 10, 2026

Research Interests

Model collaboration, knowledge and factuality, social NLP, networks and structures.

Education

University of Washington. Seattle, WA, USA PhD student in Computer Science and Engineering	2022.9-present
University of Washington. Seattle, WA, USA M.S. in Computer Science and Engineering	2022.9-2024.3
Xi'an Jiaotong University. Xi'an, Shaanxi, China B.E. in Computer Science and Technology	2018.9-2022.7

Selected Publications (= indicates equal contribution)

Scaling Participation in Modular AI Systems

Shangbin Feng, Yike Wang, Weijia Shi, Luke Zettlemoyer, Yejin Choi, and Yulia Tsvetkov.

In *arxiv*

MoCo: A One-Stop Shop for Model Collaboration Research

Shangbin Feng =, Yuyang Bai =, Ziyuan Yang =, Yike Wang, Zhaoxuan Tan, Jiajie Yan, Zhenyu Lei, Wenxuan Ding, Weijia Shi, Haojin Wang, Zhenting Qi, Yuru Jiang, Heng Wang, Chengsong Huang, Yu Fei, Jihan Yao, Yilun Du, Luke Zettlemoyer, Yejin Choi, and Yulia Tsvetkov.

In *arxiv*

When One LLM Drools, Multi-LLM Collaboration Rules

Shangbin Feng, Wenxuan Ding, Alisa Liu, Zifeng Wang, Weijia Shi, Yike Wang, Zejiang Shen, Xiaochuang Han, Hunter Lang, Chen-Yu Lee, Tomas Pfister, Yejin Choi, and Yulia Tsvetkov.

In *Proceedings of ACL 2026*

Model Swarms: Collaborative Search to Adapt LLM Experts via Swarm Intelligence

Shangbin Feng, Zifeng Wang, Yike Wang, Sayna Ebrahimi, Hamid Palangi, Lesly Miculicich, Achin Kulshrestha, Nathalie Rauschmayr, Yejin Choi, Yulia Tsvetkov, Chen-Yu Lee, and Tomas Pfister.

In *Proceedings of ICML 2025*

Don't Hallucinate, Abstain: Identifying LLM Knowledge Gaps via Multi-LLM Collaboration

Shangbin Feng, Weijia Shi, Yike Wang, Wenxuan Ding, Vidhisha Balachandran, and Yulia Tsvetkov.

In *Proceedings of ACL 2024*

Outstanding Paper Award (top 1%);

Knowledge Card: Filling LLMs' Knowledge Gaps with Plug-in Specialized Language Models

Shangbin Feng, Weijia Shi, Yuyang Bai, Vidhisha Balachandran, Tianxing He, and Yulia Tsvetkov.

In *Proceedings of ICLR 2024*

Oral (top 1.2%)

Can Language Models Solve Graph Problems in Natural Language?

Heng Wang=, Shangbin Feng==, Tianxing He, Zhaoxuan Tan, Xiaochuang Han, and Yulia Tsvetkov.

In *Proceedings of NeurIPS 2023*

Spotlight (top 3.4%)

From Pretraining Data to Language Models to Downstream Tasks: Tracking the Trails of Political Biases Leading to Unfair NLP Models

Shangbin Feng, Chan Young Park, Yuhan Liu, and Yulia Tsvetkov.

In *Proceedings of ACL 2023*

Best Paper Award (3/4864)

[[Washington Post article](#)] [[MIT Technology Review article](#)]

Publications

2026

Scaling Participation in Modular AI Systems

Shangbin Feng, Yike Wang, Weijia Shi, Luke Zettlemoyer, Yejin Choi, and Yulia Tsvetkov.

In *arxiv*

Among Us: Measuring and Mitigating Malicious Contributions in Model Collaboration Systems

Ziyuan Yang=, Wenxuan Ding=, Shangbin Feng==, and Yulia Tsvetkov.

In *Proceedings of ACL 2026*, **SAC Highlight Award**

When One LLM Drools, Multi-LLM Collaboration Rules

Shangbin Feng, Wenxuan Ding, Alisa Liu, Zifeng Wang, Weijia Shi, Yike Wang, Zejiang Shen, Xiaochuang Han, Hunter Lang, Chen-Yu Lee, Tomas Pfister, Yejin Choi, and Yulia Tsvetkov.

In *Proceedings of ACL 2026*

Data Swarms: Optimizable Generation of Synthetic Evaluation Data

Shangbin Feng, Yike Wang, Weijia Shi, and Yulia Tsvetkov.

In *Proceedings of ACL 2026, findings*

Generalizable LLM Learning of Graph Synthetic Data with Reinforcement Learning

Yizhuo Zhang=, Heng Wang=, Shangbin Feng==, Zhaoxuan Tan, Xinyun Liu, and Yulia Tsvetkov.

In *Proceedings of ACL 2026, findings*

MoCo: A One-Stop Shop for Model Collaboration Research

Shangbin Feng==, Yuyang Bai=, Ziyuan Yang=, Yike Wang, Zhaoxuan Tan, Jiajie Yan, Zhenyu Lei, Wenxuan Ding, Weijia Shi, Haojin Wang, Zhenting Qi, Yuru Jiang, Heng Wang, Chengsong Huang, Yu Fei, Jihan Yao, Yilun Du, Luke Zettlemoyer, Yejin Choi, and Yulia Tsvetkov.

In *arxiv*

Small Reward Models via Backward Inference

Yike Wang, Faeze Brahman, [Shangbin Feng](#), Teng Xiao, Hannaneh Hajishirzi, and Yulia Tsvetkov.

In *arxiv*

MentorCollab: Selective Large-to-Small Inference-Time Guidance for Efficient Reasoning

Haojin Wang, Yike Wang, [Shangbin Feng](#), Hannaneh Hajishirzi, and Yulia Tsvetkov.

In *arxiv*

Interactive Reasoning: Visualizing and Controlling Chain-of-Thought Reasoning in Large Language Models

Rock Yuren Pang, K. J. Kevin Feng, [Shangbin Feng](#), Chu Li, Weijia Shi, Yulia Tsvetkov, Jeffrey Heer, and Katharina Reinecke.

In *Proceedings of IUI 2026*

2025

Sparta Alignment: Collectively Aligning Multiple Language Models through Combat

Yuru Jiang=, Wenxuan Ding=, [Shangbin Feng=](#), Greg Durrett, and Yulia Tsvetkov.

In *Proceedings of NeurIPS 2025*

Heterogeneous Swarms: Jointly Optimizing Model Roles and Weights for Multi-LLM Systems

[Shangbin Feng](#), Zifeng Wang, Palash Goyal, Yike Wang, Weijia Shi, Huang Xia, Hamid Palangi, Luke Zettlemoyer, Yulia Tsvetkov, Chen-Yu Lee, and Tomas Pfister.

In *Proceedings of NeurIPS 2025*

Escaping the SpuriVerse: Can Large Vision-Language Models Generalize Beyond Seen Spurious Correlations?

Yiwei Yang, Chung Peng Lee, [Shangbin Feng](#), Dora Zhao, Bingbing Wen, Anthony Z. Liu, Yulia Tsvetkov, and Bill Howe.

In *Proceedings of NeurIPS 2025, Datasets and Benchmarks Track*

GuessBench: Sensemaking Multimodal Creativity in the Wild

Zifeng Zhu=, [Shangbin Feng=](#), Herun Wan, Ningnan Wang, Minnan Luo, and Yulia Tsvetkov.

In *arxiv*

ScienceMeter: Tracking Scientific Knowledge Updates in Language Models

Yike Wang, [Shangbin Feng](#), Yulia Tsvetkov, and Hannaneh Hajishirzi.

In *arxiv*

MARVEL: Modular Abstention for Reliable and Versatile Expert LLMs

Bingbing Wen, Faeze Brahman, Zhan Su, [Shangbin Feng](#), Yulia Tsvetkov, Lucy Lu Wang, and Bill Howe.

In *arxiv*

MMM: a Comprehensive and Reliable Evaluation Suite for Multitask Multimodal Generation

Jihan Yao, Yushi Hu, Yujie Yi, Bin Han, [Shangbin Feng](#), Guang Yang, Bingbing Wen, Ranjay Krishna, Lucy Lu Wang, Yulia Tsvetkov, Noah A. Smith, and Banghua Zhu.

In *arxiv*

Model Swarms: Collaborative Search to Adapt LLM Experts via Swarm Intelligence

Shangbin Feng, Zifeng Wang, Yike Wang, Sayna Ebrahimi, Hamid Palangi, Lesly Miculicich, Achin Kulshrestha, Nathalie Rauschmayr, Yejin Choi, Yulia Tsvetkov, Chen-Yu Lee, and Tomas Pfister.
In *Proceedings of ICML 2025*

Political Neutrality in AI is Impossible-But Here is How to Approximate it

Jillian Fisher, Ruth E Appel, Chan Young Park, Yujin Potter, Liwei Jiang, Taylor Sorensen, Shangbin Feng, Yulia Tsvetkov, Margaret E Roberts, Jennifer Pan, Dawn Song, and Yejin Choi.
In *Proceedings of ICML 2025, Position Track*

Oral

Biased AI can Influence Political Decision-Making

Jillian Fisher, Shangbin Feng, Robert Aron, Thomas Richardson, Yejin Choi, Daniel W. Fisher, Jennifer Pan, Yulia Tsvetkov, and Katharina Reinecke.
In *Proceedings of ACL 2025*

CodeTaxo: Enhancing Taxonomy Expansion with Limited Examples via Code Language Prompts

Qingkai Zeng, Yuyang Bai, Zhaoxuan Tan, Zhenyu Wu, Shangbin Feng, and Meng Jiang.
In *Proceedings of ACL 2025, findings*

FACTS&EVIDENCE: An Interactive Tool for Transparent Fine-Grained Factual Verification of Machine-Generated Text

Varich Boonsanong, Vidhisha Balachandran, Xiaochuang Han, Shangbin Feng, Lucy Lu Wang, and Yulia Tsvetkov.
In *Proceedings of NAACL 2025, Demo Track*

Varying Shades of Wrong: Aligning LLMs with Wrong Answers Only

Jihan Yao=, Wenxuan Ding=, Shangbin Feng=, Lucy Lu Wang, and Yulia Tsvetkov.
In *Proceedings of ICLR 2025*

2024

Know Your Limits: A Survey of Abstention in Large Language Models

Bingbing Wen, Jihan Yao, Shangbin Feng, Chenjun Xu, Yulia Tsvetkov, Bill Howe, and Lucy Lu Wang.
In *Proceedings of TACL 2024*

MEDIQ: Question-Asking LLMs for Adaptive and Reliable Clinical Reasoning

Shuyue Stella Li, Vidhisha Balachandran, Shangbin Feng, Jonathan Ilgen, Emma Pierson, Pang Wei Koh, and Yulia Tsvetkov.
In *Proceedings of NeurIPS 2024*

Teaching LLMs to Abstain across Languages via Multilingual Feedback

Shangbin Feng, Weijia Shi, Yike Wang, Wenxuan Ding, Orevaoghene Ahia, Shuyue Stella Li, Vidhisha Balachandran, Sunayana Sitaram, and Yulia Tsvetkov.
In *Proceedings of EMNLP 2024*

Modular Pluralism: Pluralistic Alignment via Multi-LLM Collaboration

Shangbin Feng, Taylor Sorensen, Yuhan Liu, Jillian Fisher, Chan Young Park, Yejin Choi, and Yulia Tsvetkov.

In *Proceedings of EMNLP 2024*

Can LLM Graph Reasoning Generalize beyond Pattern Memorization?

Yizhuo Zhang=, Heng Wang=, Shangbin Feng=, Zhaoxuan Tan, Xiaochuang Han, Tianxing He, and Yulia Tsvetkov.

In *Proceedings of EMNLP 2024, findings*

Chain-of-Layer: Iteratively Prompting Large Language Models for Taxonomy Induction from Limited Examples

Qingkai Zeng=, Yuyang Bai=, Zhaoxuan Tan, Shangbin Feng, Zhenwen Liang, Zhihan Zhang, and Meng Jiang.

In *Proceedings of CIKM 2024*

Resolving Knowledge Conflicts in Large Language Models

Yike Wang=, Shangbin Feng=, Heng Wang, Weijia Shi, Vidhisha Balachandran, Tianxing He, and Yulia Tsvetkov.

In *Proceedings of COLM 2024*

Don't Hallucinate, Abstain: Identifying LLM Knowledge Gaps via Multi-LLM Collaboration

Shangbin Feng, Weijia Shi, Yike Wang, Wenxuan Ding, Vidhisha Balachandran, and Yulia Tsvetkov.

In *Proceedings of ACL 2024*

Outstanding Paper Award (top 1%); Area Chair Award, QA Track (1/~200)

What Does the Bot Say? Opportunities and Risks of Large Language Models in Social Media Bot Detection

Shangbin Feng, Herun Wan, Ningnan Wang, Zhaoxuan Tan, Minnan Luo, and Yulia Tsvetkov.

In *Proceedings of ACL 2024*

Stumbling Blocks: Stress Testing the Robustness of Machine-Generated Text Detectors Under Attacks

Yichen Wang, Shangbin Feng, Abe Bohan Hou, Xiao Pu, Chao Shen, Xiaoming Liu, Yulia Tsvetkov, and Tianxing He.

In *Proceedings of ACL 2024*

Knowledge Crosswords: Geometric Reasoning over Structured Knowledge with Large Language Models

Wenxuan Ding=, Shangbin Feng=, Yuhao Liu, Zhaoxuan Tan, Vidhisha Balachandran, Tianxing He, and Yulia Tsvetkov.

In *Proceedings of ACL 2024, findings*

DELL: Generating Reactions and Explanations for LLM-Based Misinformation Detection

Herun Wan=, Shangbin Feng=, Zhaoxuan Tan, Heng Wang, Yulia Tsvetkov, and Minnan Luo.

In *Proceedings of ACL 2024, findings*

P³SUM: Preserving Author's Perspective in News Summarization with Diffusion Language Models

Yuhao Liu=, Shangbin Feng=, Xiaochuang Han, Vidhisha Balachandran, Chan Young Park, Sachin Kumar, and Yulia Tsvetkov.

In *Proceedings of NAACL 2024*

KGQuiz: Evaluating the Generalization of Encoded Knowledge in Large Language Models

Yuyang Bai=, Shangbin Feng=, Vidhisha Balachandran, Zhaoxuan Tan, Shiqi Lou, Tianxing He, and Yulia Tsvetkov.

In *Proceedings of The Web Conference 2024*

Knowledge Card: Filling LLMs' Knowledge Gaps with Plug-in Specialized Language Models

Shangbin Feng, Weijia Shi, Yuyang Bai, Vidhisha Balachandran, Tianxing He, and Yulia Tsvetkov.

In *Proceedings of ICLR 2024*

Oral (top 1.2%)

2023

FactKB: Generalizable Factuality Evaluation using Language Models Enhanced with Factual Knowledge

Shangbin Feng, Vidhisha Balachandran, Yuyang Bai, and Yulia Tsvetkov.

In *Proceedings of EMNLP 2023*

BotPercent: Estimating Bot Populations in Twitter Communities

Zhaoxuan Tan=, Shangbin Feng=, Melanie Sclar, Herun Wan, Minnan Luo, Yejin Choi, and Yulia Tsvetkov.

In *Proceedings of EMNLP 2023, findings*

Detecting Spoilers in Movie Reviews with External Movie Knowledge and User Networks

Heng Wang, Wenqian Zhang, Yuyang Bai, Zhaoxuan Tan, Shangbin Feng, Qinghua Zheng, and Minnan Luo.

In *Proceedings of EMNLP 2023*

Can Language Models Solve Graph Problems in Natural Language?

Heng Wang=, Shangbin Feng=, Tianxing He, Zhaoxuan Tan, Xiaochuang Han, and Yulia Tsvetkov.

In *Proceedings of NeurIPS 2023*

Spotlight (top 3.4%)

From Pretraining Data to Language Models to Downstream Tasks: Tracking the Trails of Political Biases Leading to Unfair NLP Models

Shangbin Feng, Chan Young Park, Yuhan Liu, and Yulia Tsvetkov.

In *Proceedings of ACL 2023*

Best Paper Award (3/4864)

KALM: Knowledge-Aware Integration of Local, Document, and Global Contexts for Long Document Understanding

Shangbin Feng, Zhaoxuan Tan, Wenqian Zhang, Zhenyu Lei, and Yulia Tsvetkov.

In *Proceedings of ACL 2023*

BIC: Twitter Bot Detection with Text-Graph Interaction and Semantic Consistency

Zhenyu Lei=, Herun Wan=, Wenqian Zhang, Shangbin Feng, Zilong Chen, Jundong Li, Qinghua Zheng, and Minnan Luo.

In *Proceedings of ACL 2023*

BotMoE: Twitter Bot Detection with Community-Aware Mixtures of Modal-Specific Experts

Yuhan Liu, Zhaoxuan Tan, Heng Wang, Shangbin Feng, Qinghua Zheng, and Minnan Luo.

In *Proceedings of SIGIR 2023*

AHEAD: A Triple Attention Based Heterogeneous Graph Anomaly Detection Approach

Shujie Yang, Binchi Zhang, Shangbin Feng, Zhaoxuan Tan, Qinghua Zheng, Jun Zhou, and Minnan Luo.

In *Proceedings of CIAC 2023*

KRACL: Contrastive Learning with Graph Context Modeling for Sparse Knowledge Graph Completion

Zhaoxuan Tan, Zilong Chen, Shangbin Feng, Qingyue Zhang, Qinghua Zheng, Jundong Li, and Minnan Luo.

In *Proceedings of The Web Conference 2023*

2022

PAR: Political Actor Representation Learning with Social Context and Expert Knowledge

Shangbin Feng, Zhaoxuan Tan, Zilong Chen, Peisheng Yu, Ningnan Wang, Qinghua Zheng, Xiaojun Chang, and Minnan Luo.

In *Proceedings of EMNLP 2022*

Twibot-22: Towards Graph-Based Twitter Bot Detection.

Shangbin Feng =, Zhaoxuan Tan =, Herun Wan =, Ningnan Wang =, Zilong Chen =, Binchi Zhang =, Qinghua Zheng, Wenqian Zhang, Zhenyu Lei, Shujie Yang, Xinshun Feng, Qingyue Zhang, Hongrui Wang, Yuhan Liu, Yuyang Bai, Heng Wang, Zijian Cai, Yanbo Wang, Lijing Zheng, Zihan Ma, Jundong Li, and Minnan Luo.

In *Proceedings of NeurIPS 2022, Datasets and Benchmarks Track*

GraTo: Graph Neural Network Framework Tackling Over-Smoothing with Neural Architecture Search.

Xinshun Feng, Herun Wan, Shangbin Feng, Hongrui Wang, Qinghua Zheng, Jun Zhou, and Minnan Luo.

In *Proceedings of CIKM 2022*

Datavoidant: An AI System for Addressing Political Data Voids on Social Media.

Claudia Flores-Saviaga, Shangbin Feng, and Saiph Savage.

In *Proceedings of CSCW 2022*

KCD: Knowledge Walks and Textual Cues Enhanced Political Perspective Detection in News Media.

Wenqian Zhang =, Shangbin Feng =, Zilong Chen =, Zhenyu Lei, Jundong Li, and Minnan Luo.

In *Proceedings of NAACL 2022*

Heterogeneity-aware Twitter Bot Detection with Relational Graph Transformers.

Shangbin Feng, Zhaoxuan Tan, Rui Li, and Minnan Luo.

In *Proceedings of AAAI 2022*

2021

Knowledge Graph Augmented Political Perspective Detection in News Media.

Shangbin Feng =, Zilong Chen =, Wenqian Zhang =, Qingyao Li, Qinghua Zheng, Xiaojun Chang, and

Minnan Luo.

In *arxiv*

PPSGCN: A Privacy-Preserving Subgraph Sampling Based Distributed GCN Training Method.

Binchi Zhang, Minnan Luo, Shangbin Feng, Ziqi Liu, Jun Zhou, and Qinghua Zheng.

In *arxiv*

BotRGCN: Twitter Bot Detection with Relational Graph Convolutional Networks.

Shangbin Feng, Herun Wan, Ningnan Wang, and Minnan Luo.

In *Proceedings of ASONAM 2021, Short Paper*

Twibot-20: A Comprehensive Twitter Bot Detection Benchmark.

Shangbin Feng, Herun Wan, Ningnan Wang, Jundong Li, and Minnan Luo.

In *Proceedings of CIKM 2021, Resource Track*

SATAR: A Self-supervised Approach to Twitter Account Representation Learning and its Application in Bot Detection.

Shangbin Feng, Herun Wan, Ningnan Wang, Jundong Li, and Minnan Luo.

In *Proceedings of CIKM 2021, Applied Track*

Research Internships

Research Intern, NVIDIA, Seattle, WA, USA

2026.6-2024.9

Host: David Acuna

Student Researcher, Google, Los Angeles, CA, USA

2024.6-2024.9

Host: Zifeng Wang, Chen-Yu Lee

Honors and Awards

NVIDIA Graduate Fellowship

2026-2027 school year

Jane Street Graduate Research Fellowship

2025-2026 school year

IBM PhD Fellowship

2024-2025 school year

Outstanding Reviewer, *EMNLP 2024*

2024

Outstanding Paper Award, *ACL 2024*

2024

Area Chair Award, QA Track, *ACL 2024*

2024

Best Paper Award, *ACL 2023*

2023

Top Reviewer, *Leaning on Graphs Conference 2022*

2022

Top Reviewer, *NeurIPS 2022*

2022

Service

Action Editor for *ACL Rolling Review*

from 2024

Reviewer for *TKDE*

from 2025

Reviewer for *AISTATS*

from 2025

Reviewer for *COLING*

from 2025

Reviewer for *AAAI*

from 2025

Reviewer for *IJCAI*

from 2024

Reviewer for <i>COLM</i>	from 2024
Reviewer for <i>WebConf</i>	from 2024
Reviewer for <i>EMNLP</i>	from 2023
Reviewer for <i>ACL</i>	from 2023
Reviewer for <i>ICCV</i>	from 2023
Reviewer for <i>ICML</i>	from 2022
Reviewer for <i>CVPR</i>	from 2023
Reviewer for <i>ICLR</i>	from 2023
Reviewer for <i>ICWSM</i>	from 2023
Reviewer for <i>ACL Rolling Review</i>	from 2022
Reviewer for <i>Learning on Graphs Conference</i>	from 2022
Reviewer for <i>NeurIPS, Datasets and Benchmarks Track</i>	from 2022
Reviewer for <i>ECCV</i>	from 2022
Reviewer for <i>NeurIPS</i>	from 2022
Reviewer for <i>CSCW</i>	2022
Reviewer for <i>Social Network Analysis and Mining</i>	2021

Talks

Protocols of Model Collaboration

Notre Dame, UIUC, Northwestern, Stanford NLP Seminar. February, 2026.

UPenn, Harvard, MIT, Northeastern, Cornell Tech, Columbia. November, 2025.

USC, UCLA, SaharaAI. October, 2025.

Model Swarms: Collaborative Search to Adapt LLM Experts via Swarm Intelligence

Jane Street Graduate Research Fellowship Workshop. May, 2025.

Google Deepmind. April, 2025.

Coding Aperitivo Seminar, Bocconi University. January, 2025.

From Pretraining Data to Language Models to Downstream Tasks: Tracking the Trails of Political Biases Leading to Unfair NLP Models

ACL 2023 Best Paper Award Plenary Session. July, 2023.

2023 NYU Disinformation Symposium. June, 2023.

Selected Media Coverage

ChatGPT leans liberal, research shows.

Washington Post. Gerrit De Vynck. Aug 16, 2023.

AI language models are rife with different political biases.

MIT Technology Review. Melissa Heikkilä. Aug 7, 2023.